

SKILLWEEK12

TASK1: To Execute Large Pipelines, Launch Multiple Jobs, Measure Stability, Monitor Resources, Detect Failures, Analyze Performance

```
jhansi_priya@Jhansi-Priya: ~/ x + v
jhansi_priya@Jhansi-Priya:~$ mkdir Skill12
jhansi_priya@Jhansi-Priya:~$ cd Skill12
jhansi_priya@Jhansi-Priya:~/Skill12$ ls -l | grep ".txt" | wc -l
0
jhansi_priya@Jhansi-Priya:~/Skill12$ ps aux | grep python | wc -l
1
2
jhansi_priya@Jhansi-Priya:~/Skill12$ sleep 10 &
sleep 15 &
sleep 20 &
[1] 519
[2] 520
[3] 521
jhansi_priya@Jhansi-Priya:~/Skill12$ jobs
[1]  Running                sleep 10 &
[2]-  Running                sleep 15 &
[3]+  Running                sleep 20 &
jhansi_priya@Jhansi-Priya:~/Skill12$ ps
  PID TTY          TIME CMD
 385 pts/0    00:00:00 bash
 521 pts/0    00:00:00 sleep
 522 pts/0    00:00:00 ps
[1]  Done                sleep 10
[2]-  Done                sleep 15
jhansi_priya@Jhansi-Priya:~/Skill12$ echo "Job 1 started"; sleep
5; echo "Job 1 finished" &
echo "Job 2 started"; sleep 8; echo "Job 2 finished" &
echo "Job 3 started"; sleep 10; echo "Job 3 finished" &
Job 1 started
[3]+  Done                sleep 20
[1] 524
Job 1 finished
```

```
[1] 524
Job 1 finished
Job 2 started
[1]+  Done                echo "Job 1 finished"
[1] 526
Job 2 finished
Job 3 started
[1]+  Done                echo "Job 2 finished"
[1] 530
Job 3 finished
jhansi_priya@Jhansi-Priya:~/Skill12$ jobs
[1]+  Done                echo "Job 3 finished"
jhansi_priya@Jhansi-Priya:~/Skill12$ for i in {1..10}
do
    echo "Run $i"
    date
    sleep 2
done
Run 1
Sun Sep 27 08:10:57 UTC 2026
Run 2
Sun Sep 27 08:10:59 UTC 2026
Run 3
Sun Sep 27 08:11:01 UTC 2026
Run 4
Sun Sep 27 08:11:03 UTC 2026
Run 5
Sun Sep 27 08:11:05 UTC 2026
Run 6
Sun Sep 27 08:11:07 UTC 2026
Run 7
Sun Sep 27 08:11:09 UTC 2026
```

```
jhansi_priya@Jhansi-Priya: ~/ X + v
Sun Sep 27 08:11:16 UTC 2026
jhansi_priya@Jhansi-Priya:~/Skill12$ for i in {1..4}
do
    echo "Run $i"
    date
done > stability.log
jhansi_priya@Jhansi-Priya:~/Skill12$ cat stability.log
Run 1
Sun Sep 27 08:11:39 UTC 2026
Run 2
Sun Sep 27 08:11:39 UTC 2026
Run 3
Sun Sep 27 08:11:39 UTC 2026
Run 4
Sun Sep 27 08:11:39 UTC 2026
jhansi_priya@Jhansi-Priya:~/Skill12$ top
top - 08:12:01 up 3 min, 1 user, load average: 0.10, 0.07, 0.
Tasks: 31 total, 1 running, 30 sleeping, 0 stopped, 0 z
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi,
MiB Mem : 7635.6 total, 5854.9 free, 512.9 used, 1418.
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7122.

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM
   566 jhansi_+  20   0    9292    5732   3568 R   0.3   0.1
     1 root      20   0   21704   13076   9692 S   0.0   0.2
     2 root      20   0    3180    2208   2072 S   0.0   0.0
     7 root      20   0    3196    2124   2008 S   0.0   0.0
    53 root      19  -1   42120   16276  15172 S   0.0   0.2
    98 root      20   0   25552    7000   5220 S   0.0   0.1
   111 root      20   0  152948    1792   1544 S   0.0   0.0
   114 root      20   0  153080    1864   1540 S   0.0   0.0
   119 root      20   0  153080    1916   1520 S   0.0   0.0
```

```
GNU nano 7.2 docs/
# Project Architecture

## Components
1. User
2. Application
3. Process Manager
4. Operating System
5. File System

## Execution Flow
User
|
v
Application
|
v
Process
|
v
System Call
|
v
Linux Kernel
|
v
Hardware
```

```
GNU nano 7.2 docs/diag:
+-----+
| Start |
+-----+
|
v
+-----+
| Read Data |
+-----+
|
v
+-----+
| Process Data |
+-----+
|
v
+-----+
| Check Errors |
+-----+
|
v
+-----+
| Output |
+-----+
|
v
+-----+
| End |
+-----+
```

```
GNU nano 7.2 docs/execution_path.
# Execution Path

1. User starts the application.
2. Linux creates a process.
3. The process executes instructions.
4. The program requests resources from Linux.
5. Linux handles the system call.
6. The program receives the result.
7. The program produces output.
8. The process terminates.
```

```
GNU nano 7.2 docs/design/design_decisions.md
# Design Decisions

## Decision 1
We use Linux processes to execute independent jobs.

## Decision 2
Pipelines are used to transfer output between commands.

## Decision 3
Logs are stored for detecting failures.

## Decision 4
System resources are monitored using Linux utilities.

## Decision 5
Git is used for source code management.
```

TASK2: To Document Architecture, Create Flow Diagrams, Explain Execution Paths, Describe Syscalls, Record Design Decisions, Review Documentation

```
jhansi_priya@Jhansi-Priya: ~/ / + ^ v
jhansi_priya@Jhansi-Priya:~$ cd Skill12
jhansi_priya@Jhansi-Priya:~/Skill12$ mkdir docs
mkdir docs/diagrams
mkdir docs/design
jhansi_priya@Jhansi-Priya:~/Skill12$ tree
├── docs
│   ├── design
│   ├── diagrams
│   └── stability.log
4 directories, 1 file
jhansi_priya@Jhansi-Priya:~/Skill12$ nano docs/architecture.md
jhansi_priya@Jhansi-Priya:~/Skill12$ nano docs/diagrams/flow.txt
jhansi_priya@Jhansi-Priya:~/Skill12$ nano docs/execution_path.md
jhansi_priya@Jhansi-Priya:~/Skill12$ sudo apt update
[sudo] password for jhansi_priya:
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/7.0 InRelease
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Get:6 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1319 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1697
kB]
Fetched 3142 kB in 3s (1159 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
79 packages can be upgraded. Run 'apt list --upgradable' to see them.
jhansi_priya@Jhansi-Priya:~/Skill12$ sudo apt install strace
Reading package lists... Done
```

```
jghansi_priya@Jghansi-Priya:~/Skill12$ strace ls  
execve("/usr/bin/ls", ["ls"], 0x7ffdle6a2f70 / * 27 vars */) = 0  
brk(NULL)                               = 0x6155625fb000  
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1,  
f130000)  
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or d  
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3  
fstat(3, {st_mode=S_IFREG|0644, st_size=25719, ...}) = 0  
mmap(NULL, 25719, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7cc7af129000  
close(3)                                = 0  
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_C  
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\  
= 832  
fstat(3, {st_mode=S_IFREG|0644, st_size=174472, ...}) = 0  
mmap(NULL, 181960, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7cc
```

```
jhansi_priya@Jhansi-Priya:~/Skill12$ nano docs/design/design_decisions.md
jhansi_priya@Jhansi-Priya:~/Skill12$ find docs -type f
docs/architecture.md
docs/diagrams/flow.txt
docs/execution_path.md
docs/design/design_decisions.md
jhansi_priya@Jhansi-Priya:~/Skill12$ cat docs/architecture.md
# Project Architecture

## Components

1. User
2. Application
3. Process Manager
4. Operating System
5. File System

## Execution Flow

User
|
v
Application
|
v
Process
|
v
System Call
|
v
Linux Kernel
|
v
Hardware
jhansi_priya@Jhansi-Priya:~/Skill12$ cat docs/execution_path.md
cat docs/design/design_decisions.md
# Execution Path

1. User starts the application.
2. Linux creates a process.
3. The process executes instructions.
```

```

Hardware
jhansi_priya@Jhansi-Priya:~/Skill12$ cat docs/execution_path.md
cat docs/design/design_decisions.md
# Execution Path

1. User starts the application.
2. Linux creates a process.
3. The process executes instructions.
4. The program requests resources from Linux.
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7. The program produces output.
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# Design Decisions

## Decision 1
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## Decision 2
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## Decision 3
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System resources are monitored using Linux utilities.

## Decision 5
Git is used for source code management.
jhansi_priya@Jhansi-Priya:~/Skill12$ |

```

TASK3: To Organize Source Code, Improve Build Scripts, Add README, Verify Repository Structure, Tag Releases, Validate Deliverables

```

GNU nano 7.2 src/main.c
#include <stdio.h>

int main()
{
    printf("Project executed successfully!\n");
    return 0;
}

GNU nano 7.2 README.md *
# Linux Project

## Description
This project demonstrates Linux processes,
pipelines, system calls, monitoring and
performance analysis.

## Directory Structure
- src - Source code
- tests - Test files
- scripts - Utility scripts
- docs - Documentation
- build - Compiled files

## Build
Run:
./build.sh

## Execute
Run:
./build/main

```

```
jhansi_priya@Jhansi-Priya: ~/ × + ⌵
jhansi_priya@Jhansi-Priya:~/Skill12$ mkdir -p src
mkdir -p tests
mkdir -p scripts
mkdir -p docs
jhansi_priya@Jhansi-Priya:~/Skill12$ tree
.
├── docs
│   ├── architecture.md
│   ├── design
│   │   └── design_decisions.md
│   ├── diagrams
│   │   └── flow.txt
│   └── execution_path.md
├── scripts
├── src
├── stability.log
├── systemcalls.txt
└── tests

7 directories, 6 files
jhansi_priya@Jhansi-Priya:~/Skill12$ nano src/main.c
jhansi_priya@Jhansi-Priya:~/Skill12$ nano build.sh
jhansi_priya@Jhansi-Priya:~/Skill12$ chmod +x build.sh
jhansi_priya@Jhansi-Priya:~/Skill12$ ./build.sh
Building project...
Build successful!
jhansi_priya@Jhansi-Priya:~/Skill12$ ./build/main
Project executed successfully!
jhansi_priya@Jhansi-Priya:~/Skill12$ nano README.md
jhansi_priya@Jhansi-Priya:~/Skill12$ git --version
git version 2.43.0
jhansi_priya@Jhansi-Priya:~/Skill12$ git status
```

```
jhansi_priya@Jhansi-Priya: ~/ × + ⌵
jhansi_priya@Jhansi-Priya:~/Skill12$ git status
fatal: not a git repository (or any of the parent directories): .git
jhansi_priya@Jhansi-Priya:~/Skill12$ git init
hint: Using 'master' as the name for the initial branch. This default branch
hint: name
hint: is subject to change. To configure the initial branch name to use in
hint: all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this comman
hint: d:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/jhansi_priya/Skill12/.git/
jhansi_priya@Jhansi-Priya:~/Skill12$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    README.md
    build.sh
    build/
    docs/
    src/
    stability.log
    systemcalls.txt

nothing added to commit but untracked files present (use "git add" to track
)
jhansi_priya@Jhansi-Priya:~/Skill12$ git add .
jhansi_priya@Jhansi-Priya:~/Skill12$ git status
On branch master

No commits yet
```

```
jhansi_priya@Jhansi-Priya: ~/ × + ⌵
)
jhansi_priya@Jhansi-Priya:~/Skill12$ git add .
jhansi_priya@Jhansi-Priya:~/Skill12$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   README.md
    new file:   build.sh
    new file:   build/main
    new file:   docs/architecture.md
    new file:   docs/design/design_decisions.md
    new file:   docs/diagrams/flow.txt
    new file:   docs/execution_path.md
    new file:   src/main.c
    new file:   stability.log
    new file:   systemcalls.txt

jhansi_priya@Jhansi-Priya:~/Skill12$ git commit -m "Initial project setup"
Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: empty ident name (for <jhansi_priya@Jhansi-Priya.localdomain>) not a
llowed
```

TASK4: To Create Demo Scenarios, Validate Features, Rehearse Presentation, Demonstrate Pipelines, Demonstrate Signals, Final Project Review

```
GNU nano 7.2 demo/scenari
# Demo Scenarios

## Scenario 1
Run the complete application.

## Scenario 2
Run multiple jobs simultaneously.

## Scenario 3
Demonstrate a pipeline.

## Scenario 4
Monitor CPU and memory.

## Scenario 5
Demonstrate process signals.

## Scenario 6
Show error detection.

## Scenario 7
Build and execute the final project.
```

```
jhansi_priya@Jhansi-Priya: ~/Skill12$ nano demo/scenarios.md
jhansi_priya@Jhansi-Priya: ~/Skill12$ cat /etc/passwd | grep bash | wc -l
2
jhansi_priya@Jhansi-Priya: ~/Skill12$ ps aux | grep bash
jhansi_+ 456 0.0 0.0 6084 5396 pts/1 S+ 08:09 0:00 -bash
jhansi_+ 593 0.0 0.0 6216 5460 pts/0 Ss 08:14 0:00 -bash
jhansi_+ 1153 0.0 0.0 4100 2164 pts/0 S+ 08:26 0:00 grep --c
olor=auto bash
jhansi_priya@Jhansi-Priya: ~/Skill12$ sleep 1000 &
[1] 1154
jhansi_priya@Jhansi-Priya: ~/Skill12$ jobs -l
[1]+ 1154 Running sleep 1000 &
jhansi_priya@Jhansi-Priya: ~/Skill12$ kill 1154
jhansi_priya@Jhansi-Priya: ~/Skill12$ jobs
[1]+ Terminated sleep 1000
jhansi_priya@Jhansi-Priya: ~/Skill12$ sleep 1000 &
[1] 1193
jhansi_priya@Jhansi-Priya: ~/Skill12$ jobs -l
[1]+ 1193 Running sleep 1000 &
jhansi_priya@Jhansi-Priya: ~/Skill12$ kill -STOP 1193
jhansi_priya@Jhansi-Priya: ~/Skill12$ kill -CONT 1193
jhansi_priya@Jhansi-Priya: ~/Skill12$ kill 1193
jhansi_priya@Jhansi-Priya: ~/Skill12$ ./build/main
Project executed successfully!
[1]+ Terminated sleep 1000
jhansi_priya@Jhansi-Priya: ~/Skill12$ ./build.sh
Building project...
Build successful!
jhansi_priya@Jhansi-Priya: ~/Skill12$ ps aux | grep bash
jhansi_+ 456 0.0 0.0 6084 5396 pts/1 S+ 08:09 0:00 -bash
jhansi_+ 593 0.0 0.0 6732 5988 pts/0 Ss 08:14 0:00 -bash
jhansi_+ 1207 0.0 0.0 4100 2160 pts/0 S+ 08:28 0:00 grep --c
olor=auto bash
jhansi_priya@Jhansi-Priya: ~/Skill12$ strace ./build/main
execve("./build/main", ["/build/main"], 0x7ffe472eae50 /* 27 vars */) = 0
brk(NULL) = 0x5b43bd552000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) =
0x7a6c36a77000
access("/etc/ld.so.preload", R_OK) = -1 ENOENT (No such file or direct
ory)
```

```

+++ exited with 0 +++
jhansi_priya@Jhansi-Priya:~/Skill12$ time ./build/main
Project executed successfully!

real    0m0.002s
user    0m0.002s
sys     0m0.000s
jhansi_priya@Jhansi-Priya:~/Skill12$ top
top - 08:28:46 up 19 min, 1 user, load average: 0.02, 0.01, 0.00
Tasks: 32 total, 1 running, 31 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0
MiB Mem : 7635.6 total, 6982.5 free, 530.3 used, 275.0 buff/cach
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 7105.3 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+
587 root        20   0    3200    1256   1108 S   0.3   0.0   0:00.09
1 root        20   0   21788   13092   9692 S   0.0   0.2   0:00.67
2 root        20   0    3180    2208   2072 S   0.0   0.0   0:00.01
7 root        20   0    3196    2124   2008 S   0.0   0.0   0:00.00
53 root       19  -1   50448   16384   15248 S   0.0   0.2   0:00.23
98 root        20   0   25552    7000   5220 S   0.0   0.1   0:00.33
111 root       20   0  152948   1792   1544 S   0.0   0.0   0:00.00
114 root       20   0  153080   1864   1540 S   0.0   0.0   0:00.00
119 root       20   0  153080   1916   1520 S   0.0   0.0   0:00.00
126 root       20   0  153080   1688   1424 S   0.0   0.0   0:00.00
127 root       20   0  153080   1644   1384 S   0.0   0.0   0:00.00
136 root       20   0  153080   1680   1424 S   0.0   0.0   0:00.00
141 root       20   0  152948   1764   1532 S   0.0   0.0   0:00.00
145 root       20   0  452004  10888   1536 S   0.0   0.1   0:02.62
213 systemd+   20   0   21352  13228  11100 S   0.0   0.2   0:00.12
217 systemd+   20   0   91044   8048   7080 S   0.0   0.1   0:00.10
223 root       20   0    4248    2712   2460 S   0.0   0.0   0:00.00

```

```

jhansi_priya@Jhansi-Priya:~/Skill12$ ./build/main
Project executed successfully!
jhansi_priya@Jhansi-Priya:~/Skill12$ find src -type f
src/main.c
jhansi_priya@Jhansi-Priya:~/Skill12$ find docs -type f
docs/architecture.md
docs/diagrams/flow.txt
docs/execution_path.md
docs/design/design_decisions.md
jhansi_priya@Jhansi-Priya:~/Skill12$

```


